

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) has emerged as a general and increasingly influential framework for strengthening the reasoning capabilities of large language models across a broad range of tasks. Yet a recurring obstacle limits the family of policy optimization algorithms it relies on: methods such as GRPO frequently produce zero gradients. This failure mode arises primarily from two design choices that are shared widely across such algorithms—fixed clipping bounds imposed uniformly on token-level probability ratios, and the standardization of identical rewards—both of which tend to yield ineffective gradient updates and leave much of the generated data underexploited. In this work, we propose **Dynamic Clipping Policy Optimization (DCPO)**, a general-purpose approach built on two complementary ideas: a dynamic clipping strategy that adaptively reshapes clipping bounds according to token-specific prior probabilities so as to broaden token-level exploration, and a smooth advantage standardization technique that aggregates rewards across cumulative training steps to improve the response-level utilization of generated data. Because both mechanisms are agnostic to model family and scale, they apply naturally across diverse policy-optimization settings. DCPO delivers state-of-the-art performance across four benchmarks and four different models spanning a wide range of scales and pretraining regimes. In particular, DCPO achieved an Avg@1 of 46.7 under greedy decoding and an Avg@32 of 38.8 under 32-sample decoding on the AIME24 benchmark, surpassing DAPO (36.7/31.6), GRPO (36.7/32.1) and GSPO (40.0/34.9) on the Qwen2.5-Math-7B model. On the AIME25 benchmark with Qwen2.5-14B, DCPO reaches (23.3/19.0), surpassing GRPO (13.3/10.5), DAPO (20.0/15.3) and GSPO (16.7/9.9). Beyond raw accuracy, DCPO achieved on average a 28% improvement in the nonzero-advantage ratio over GRPO across all four models, doubled training efficiency relative to DAPO, and reduced the token clipping ratio by an order of magnitude compared to both GRPO and DAPO, while consistently achieving superior performance. Taken together, these results indicate that DCPO offers a broadly effective and transferable recipe for leveraging generated data more efficiently in reinforcement learning of large language models.

1 INTRODUCTION

In recent years, the reasoning capabilities of large language models (LLMs) have attracted growing attention across the research community (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Among the techniques proposed to strengthen these capabilities, Reinforcement Learning from Verifiable Rewards (RLVR) has emerged as a particularly general and promising one. By leveraging rule-based outcome rewards to optimize LLMs, it improves performance across a wide range of domains that demand reflective, multi-step decision-making—mathematics and coding being two prominent examples, but by no means the only settings in which the paradigm applies (DeepSeek-AI et al., 2024).

To operationalize RLVR at scale, the GRPO (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) family of algorithms was developed and has since been adopted broadly. Although the robustness of GRPO has been extensively validated across many studies and settings, more recent work such as DAPO (Yu et al., 2025) has surfaced limitations that appear to be shared across this family of methods rather than confined to any single implementation. In particular, GRPO is prone to entropy collapse, in which advantage values decay toward zero during the later stages of

training. Compounding this, poorly matched weight-clipping tends to discard rarely sampled learning signals, systematically narrowing the diversity of model exploration.

A variety of remedies have been proposed. DAPO, for instance, introduces the Clip-Higher strategy to counter entropy collapse by relaxing the upper clipping bound, together with a Dynamic Sampling strategy that filters out samples with uniform rewards so that every batch retains only samples carrying valid gradients. Such fixes, however, tend to share two limitations that recur across the broader landscape of policy-optimization methods: (i) **Response-Level Inefficiency**: dynamic sampling inherently lowers sampling efficiency and thereby slows convergence. (ii) **Token Clipping Deficit**: our analysis indicates that the optimal clipping bound is not a single global constant but varies from token to token, so any uniform upper bound is suboptimal. Even approaches that avoid discarding zero-gradient responses outright do not escape these underlying issues, which suggests the limitations are structural rather than incidental to a particular algorithm.

In this work, we introduce **Dynamic Clipping Policy Optimization (DCPO)**, a general approach designed to address these limitations at their root and, in doing so, to apply broadly across models and policy-optimization settings. First, we introduce a **smoothing technique that standardizes advantages across cumulative steps**, targeting the gradient-vanishing problem that arises when the advantages of equally rewarded responses are zeroed out. By aggregating the reward distribution of all cumulatively generated responses with the current generation distribution, the method enables more efficient response-level exploration (see Section 2). Second, we propose a **dynamic clipping method that adaptively adjusts the clipping bounds based on old, token-specific probabilities**, replacing the practice of fixing upper and lower clipping bounds—whether symmetric or asymmetric—with a single probability-aware rule. Concretely, tokens with lower prior probabilities are granted wider clipping bounds, opening up more room for token-level exploration (see Section 2 and Appendix A.6). In addition, DCPO computes the mean loss only across tokens within each individual response rather than averaging over an entire batch; this preserves the relative advantage structure among responses to the same prompt and avoids the dilution effect that batch-level averaging introduces. None of these components is tied to a specific model architecture, task, or reward scheme, so the design is intended to generalize.

To evaluate the approach, we merged Math-DAPO-17k (Yu et al., 2025) with levels 3-5 of the MATH dataset (Liu et al., 2025) to form the training corpus and assessed performance on four widely used mathematical reasoning benchmarks. To probe how well the method transfers across model families and scales rather than tuning to a single configuration, we ran experiments on four different models and compared against GRPO, DAPO, and, where available, GSPO. Across this range of settings, DCPO consistently attained superior performance, with especially large gains on the challenging AIME benchmarks: on AIME24-Avg@32 under 32-sample decoding, DCPO-7B scored 38.8 vs. 32.1 (GRPO) vs. 31.6 (DAPO) vs. 34.9 (GSPO), and on AIME25-Avg@32, DCPO-14B reached 19.0 vs. 10.5 (GRPO) vs. 15.3 (DAPO) vs. 9.9 (GSPO). The advantages were not limited to accuracy: DCPO raised the average nonzero-advantage ratio by 28% over GRPO, doubled training efficiency relative to DAPO, and reduced token clipping by an order of magnitude compared to all baselines, while still delivering superior performance. The consistency of these gains across models points to benefits that hold broadly rather than in a single favorable case.

For preliminary details, including loss and advantage calculation methods for previous algorithms such as GRPO, DAPO and GSPO, see the Appendix A.2.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Clipping the probability ratio is a widely adopted and general-purpose technique in reinforcement learning, used to stabilize policy updates and prevent large, destabilizing parameter shifts during optimization. Its role becomes especially pronounced whenever importance sampling is applied in off-policy settings, a situation common to a broad class of policy-optimization algorithms. When importance sampling is reweighted by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$, one obtains an unbiased estimate of $\mathbb{E}_{q(x)}[f(x)\frac{p(x)}{q(x)}]$, yet the variance of this

estimate can be substantially inflated, as described in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

To mitigate this variance inflation, a long line of prior work (e.g., Schulman et al. (2017)) constrains the probability ratio within a fixed clipping bound governed by a hyperparameter ϵ , as in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed clipping bound, however, constrains the model’s exploration of the diversity space, and it does so most severely in regions where the old policy assigns low probability mass—a regime that arises routinely across models and tasks. In these regions the absolute bound on decreasing probabilities is inherently narrow, which limits the capacity for meaningful policy updates. Methods such as DAPO attempt to relieve this pressure through asymmetric clipping bounds, but the fundamental limitation persists across such variants. GSPO, meanwhile, discards entire responses whenever the sequence-level importance ratio exceeds the asymmetric bounds, wasting informative tokens while retaining more high-variance ones, which in turn destabilizes training.

Viewed through the lens of controlling the variance–bias trade-off, a more broadly principled approach is to impose probability-dependent constraints, as formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3) and ensuring non-negativity of the square-root term by taking its maximum with zero, we obtain a closed-form solution for $r(x)$. This yields **dynamic-adaptive clipping bounds** that adapt with $q(x)$ and the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4). Because the rule is expressed purely in terms of token probabilities, it carries over directly to any policy-optimization method that clips a probability ratio.

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

This dynamic-adaptive clipping scheme opens up greater policy exploration in low-probability regions wherever they occur. Furthermore, inspired by the dual clipping method (Ye et al., 2020), we cap the maximum bounds for both positive and negative advantages at 10, which guards against overtraining driven by excessively large clipping bounds. A full derivation and analysis of these dynamic clipping bounds within the importance-sampling framework—together with the conditions under which the bounds generalize—is provided in Appendix A.5 and Appendix A.6.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

Across the prevailing methods—GRPO, DAPO, and GSPO alike—the advantage $\hat{A}_{new,j}^i$ of the j -th response is obtained by standardizing only the rewards $R_{j=1,\dots,G}^i$ collected for the same prompt at the i -th step, as in Equation (11) in Appendix A.3. This shared design invites two problems that surface regardless of the specific algorithm. First, when randomness in response sampling drives all rewards at a given step to the same value, the advantage collapses to zero, so the prompt contributes nothing to the parameter update—even when its reasoning trajectories differ in potentially valuable ways. Second, randomness in high-entropy sampling can produce highly skewed label counts, causing standardized advantage values to fluctuate widely across steps and even reverse sign, which destabilizes training.

At the same time, recent work (Yue et al., 2025) indicates that RLVR chiefly refines a policy by adjusting the likelihood of correct samples rather than fundamentally reshaping the existing distribution—an observation that appears to hold across models and training regimes. Building on this view, the overall reward distribution of responses to the same prompt can be treated as drawn from a single global distribution throughout training, which motivates a *cumulative* standardization as in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the statistics are computed over all responses generated so far for the same prompt.

To mitigate fluctuations of both step-specific standardization ($\hat{A}_{new,j}^i$) and cumulative standardization ($\hat{A}_{total,j}^i$), we introduce two smoothing functions, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$, which represent the weighted average between the two standardization methods with the weights changing over step i in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

$\hat{S}A_{new,j}^i$ places progressively more weight on the current distribution as training advances, reflecting a preference for adapting to the latest policy outputs. Conversely, $\hat{S}A_{total,j}^i$ leans on the overall cumulative distribution, treating the deviation at the current step as a fine-tuning adjustment within the broader distribution. Our final advantage $\hat{A}_j^i$ is then taken to be whichever smoothed advantage has the smaller absolute value, a choice that dampens the fluctuations of both cumulative and step-wise standardization and thereby promotes training stability across settings, as defined in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

Once a prompt has entered the optimization process, the responses it generates in subsequent steps continue to contribute to model updates whether or not the current advantage is zero. As a result, even when identical rewards occur within a step, these responses still update the model through the advantage $\frac{1}{i} \hat{A}_{total,j}^i$, preserving useful learning signals and improving data efficiency—a benefit that applies to any prompt regardless of its difficulty.

3.1 OTM LOSS

Although the **Token-Level Mean loss (TLM)** (the loss Equation (13) used in DAPO) balances the contributions of all tokens, it disrupts the relative relationship between responses of differing lengths—an effect that appears whenever response lengths vary, as they typically do. Consider, as a representative case, two responses to the same prompt: response A with an advantage of 1 and a length of 500, and response B with an advantage of 0.5 and a length of 1500. Under TLM, the loss for A is weighted by $\frac{500}{500+1500} = 0.25$ and that for B by $\frac{1500}{500+1500} = 0.75$. The total loss contribution of A is therefore $1 \times 0.25 = 0.25$ and that of B is $0.5 \times 0.75 = 0.375$, so B exerts the larger influence (0.375) despite its lower advantage (0.5).

By contrast, averaging across batches—the **Sequence-Level Mean loss (SLM)** (the loss Equation (12) used in GRPO)—attenuates the magnitude of the advantage, effectively scaling it down by a factor of G once advantages are computed. This effect grows more damaging as the batch size increases, since it erodes the correlations encoded in the original standardized advantages. Because standardized advantages already carry meaningful relative relationships across responses, we observe that averaging over the $|o_i|$ tokens within a single response both preserves those relationships and distributes importance evenly across the tokens of that response. We therefore average only over the tokens within each individual response, without averaging over all responses in the same batch. We call this the **Only Token Mean loss (OTM)**, given in Equation (8).

$$\mathcal{T}_{DCPO}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

Following DAPO, the DCPO loss is not constrained by a KL-divergence term, and for the lower and upper clipping bounds it simply adopts the dynamic clipping bounds introduced in Section 2, so the three components combine into a single, self-contained objective.

4 EXPERIMENTAL SETUP

Models and Datasets To assess how broadly the method transfers, we experimented with four different models spanning distinct scales and pretraining regimes: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b). This selection deliberately mixes instruction-tuned and base checkpoints as well as general-purpose and math-specialized backbones, so that any consistent gains reflect properties of the method rather than of a single model type. For training we used a merged corpus of approximately 25k mathematical problems combining the DAPO-Math-17K corpus (Yu et al., 2025) with levels 3-5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). All problems were processed with the Qwen-Math template (see Section A.8) for both training and evaluation.

Evaluation We evaluated our models on four widely used mathematical reasoning benchmarks that together span a broad difficulty spectrum, from general problem sets to competition-level challenges: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). For **MATH500** we report only Avg@1, given its larger size (500 problems). For the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) sets, we report both Avg@1 and Avg@32 to obtain a robust and comprehensive picture of performance across decoding regimes.

- **Avg@1**: standard accuracy under greedy decoding, capturing the quality of the model’s single best prediction.
- **Avg@32**: the average accuracy over 32 sampled responses per problem, using a temperature of 1.0 and top_p of 1.0. This metric probes the robustness and stability of the trained policy distribution under stochastic decoding, a setting representative of how such models are deployed in practice.

Baseline Settings We adopted GRPO and DAPO as baselines throughout, and additionally included GSPO on Qwen2.5-Math-7B and Qwen2.5-14B, giving a representative cross-section of contemporary policy-optimization methods to compare against. Remaining settings—including the various clipping thresholds ϵ_* , the codebase, and other common training hyperparameters—are detailed in Appendix A.9.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1 summarizes the performance of a range of preference-optimization methods—GRPO, DAPO, GSPO, and our proposed DCPO—on four math reasoning benchmarks: MATH500, AMC23, AIME24, and AIME25. Reporting both greedy decoding (Avg@1) and sampling-based decoding (Avg@32) lets us gauge the robustness and generalization of each method across decoding regimes rather than under a single favorable one. Across the full span of model scales and pretraining types considered, DCPO consistently matches or surpasses GRPO, DAPO, and, where available, GSPO. The uniformity of this pattern—holding across small and large models, base and instruction-tuned checkpoints, and general and math-specialized backbones—indicates that DCPO functions as a broadly applicable, general-purpose preference-optimization technique for reinforcement learning of LLMs rather than a fix tuned to one configuration.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO achieves the highest average scores (47.2/32.8), exceeding GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO yields (37.6/22.7), exceeding GRPO (+1.3/+1.7) and remaining competitive with DAPO, with a clear improvement in Avg@1 (+1.0) and only a marginal gap in Avg@32 (-0.4). For Qwen2.5-Math-7B, DCPO attains (57.1/45.2), surpassing GRPO (+4.0/+3.6), DAPO (+3.3/+2.8), and GSPO (+1.9/+1.9). On the largest tested model, Qwen2.5-14B, DCPO delivers a strong (53.2/39.0), improving substantially over GRPO (+6.5/+7.7) and outperforming DAPO by (+1.3/+0.1) and GSPO by (+4.2/+5.5) in Avg@1 and Avg@32. Read together, these results show that DCPO’s gains hold consistently as model capacity grows and that its alignment to the reward signal remains superior under both greedy and stochastic decoding, evidence that the benefits generalize across scales and decoding paradigms rather than being confined to a particular one.

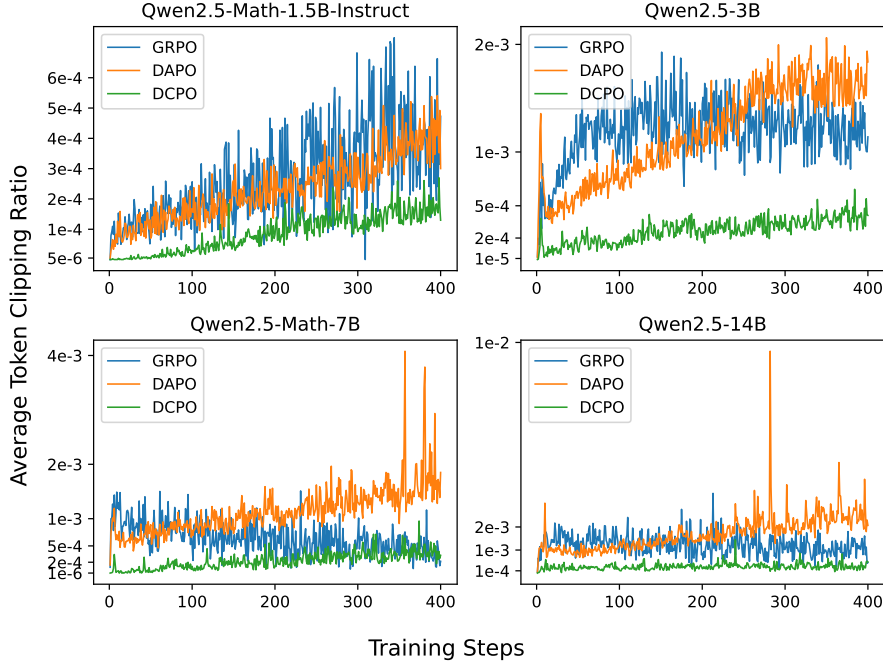
Table 1: Performance across benchmarks and across four models of varying scale and pretraining type. Avg@1: greedy decoding; Avg@32: 32-sample decoding. Boldface marks the best value in each column, and Δ reports the improvement over each baseline.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

Robustness on AIME under Sampling (Avg@32) A particularly telling trend emerges on the AIME24 and AIME25 benchmarks, high-difficulty competition math sets designed to stress-test reasoning ability under the most demanding conditions. Under Avg@32, which measures how reliably a model produces correct responses given high-entropy sampling, DCPO shows markedly stronger robustness at the 7B and 14B scales: on AIME24-Avg@32, DCPO-7B achieves 38.8, outperforming GRPO (32.1), DAPO (31.6), and GSPO (34.9); on AIME25-Avg@32, DCPO-14B reaches 19.0, exceeding GRPO (10.5), DAPO (15.3), and GSPO (9.9). These gains indicate that DCPO improves not only average-case accuracy but also the diversity and reliability of correct generations—a property that matters wherever sampling-based decoding, including beam search or temperature-driven variability, is used. Because such stochastic decoding underlies many real-world deployments in which multiple plausible outputs are drawn, DCPO’s ability to jointly optimize reward fidelity and exploration under high-entropy decoding points to broad practical value.

5.2 THE TOKEN CLIPPING RATIO (TCR)

We define the **Token Clipping Ratio (TCR)** as the proportion of tokens excluded from policy updates during back-propagation because of clipping. As Figure 1 shows, *TCR* behaves quite differently from one reinforcement learning method to another, making it a useful lens for comparing methods across

Figure 1: TCR across the full set of models and methods considered.

models.

$$TCR = Average \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } micro_m}{\text{Total number of tokens in } micro_m} \right) \quad (9)$$

where $micro_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

Under GRPO training, TCR follows divergent trends that depend on model size: for the smaller models (Qwen2.5-Math-1.5B and Qwen2.5-3B) it rises with training steps, whereas for the larger ones (Qwen2.5-Math-7B and Qwen2.5-14B) it gradually falls. DAPO, by contrast, shows a consistently upward TCR trajectory across all models, meaning that as training proceeds an ever larger share of tokens is clipped and a growing fraction of policy updates rests on partial or truncated responses. Both GRPO and DAPO, moreover, exhibit pronounced fluctuations and occasional abnormal spikes in TCR , which can inject instability into optimization and, in some cases, precipitate entropy collapse. GSPO fares no better in this respect: its average TCR exceeds 11% on Qwen2.5-Math-7B and 15% on Qwen2.5-14B, far above that of token-level clipping methods, and correspondingly wastes more informative tokens. These contrasting behaviors underscore that the clipping pathology is a general one, manifesting differently across methods and scales.

DCPO, in contrast, maintains a remarkably stable and efficient training state. Independent of model size or training stage, its TCR stays nearly constant and an order of magnitude below that of GRPO and DAPO. This lower and steadier TCR means that DCPO discards fewer tokens, allowing a greater portion of complete response sequences to inform the policy update and freeing up more room for the model to explore the diversity of rare tokens. That the effect persists uniformly across every model tested speaks to DCPO’s superior sample efficiency relative to the baselines, achieved while keeping outlier variation tightly in check.

Taking Qwen2.5-Math-7B as a representative case, we observe that the token distribution quickly concentrates on high-confidence tokens: after roughly 60 steps, about 95% of generated tokens have probability ($q(x) > 0.9$); this fraction rises to about 97% after roughly 100 steps and keeps climbing in later training. This concentration is not peculiar to one model but a general tendency of RLVR training. As illustrated in Figure 4d, once the old probability satisfies ($q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83$), the new probability $p(x)$ participates in the model update within the interval $[\frac{1}{1+\epsilon}, 1]$ without being

discarded, whatever clipping method is used. Consequently, the vast majority of tokens lie in a region that is never clipped, which accounts for the low overall *TCR* seen in practice. At the same time, Figure 1 shows that *TCR* tends to grow in the later phases of most training runs, meaning that an increasingly large share of low-probability tokens risks being discarded and the model’s ability to explore the diversity of rare tokens is progressively curtailed.

Previous work (Wang et al., 2025) has shown that **high-entropy tokens**—precisely the low-probability ones—are the primary drivers behind the emergence of reasoning capabilities in these models. Since entropy and probability are inversely related, the rarer a token, the more information it carries. Granting such low-probability tokens a broader admissible update range therefore enlarges the model’s effective exploration space and facilitates the acquisition of diverse reasoning capabilities, an argument that applies to any model trained under this paradigm.

These empirical and theoretical findings together motivate our **dynamic-adaptive clipping** scheme, which relaxes clipping for tokens with $q(x) < \frac{1}{1+\epsilon}$ while preserving the safe interval for high-confidence tokens. By adaptively widening the update interval for the high-entropy tail of the distribution, the method restores the model’s capacity to learn from rare, informative tokens and, in turn, strengthens its reasoning ability across settings.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

In RLVR training generally, a policy update depends on a given response producing a nonzero gradient. We quantify the efficiency of this process with the **Response Utilization Ratio** (*RUR*), defined as the percentage of generated responses with nonzero advantage that participate in the policy update—a measure applicable to any method in this family.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

Table 2 summarizes the average *RUR* during GRPO training, which is low—an overall mean of only 44.6% across the four different models. In other words, more than half of the generated responses are discarded, a striking inefficiency in response utilization that constrains exploration of diversity, and one that recurs across every model examined. DCPO, by contrast, reaches an average *RUR* of roughly 70% after the first epoch, an absolute gain of 28% and a relative gain of 64% over GRPO that holds broadly across the model suite.

Table 2: Average *RUR* over 400 training steps, reported for each model in the suite.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 shows that *RUR* under GRPO training declines sharply over time, and it does so regardless of model size or type—dropping from above 90% to below 50%, and on Qwen2.5-Math-7B plummeting to as low as 30% after about 200 steps. This consistent, model-agnostic decay indicates that GRPO’s reliance on current-step advantage standardization yields a large fraction of zero gradients, so its exploration of diversity becomes markedly inefficient in the later stages of training. DCPO, which instead employs a smooth advantage standardization scheme, consistently sustains a high *RUR*: after the first epoch it stabilizes around 70% and continues to improve with further training, a pattern that holds uniformly across the models studied.

We observed that *RUR* exceeded 90% in the first few steps but then settled around 70% from the second epoch through the end of training, never reaching 100%. Examining why this occurs both explains the pattern and reveals the broader value of the SAS mechanism.

First, the initial *RUR* above 90% owes largely to early-stage outputs that violate the required formatting, incurring negative rewards and inflating *RUR*. This suggests that models across the board quickly learn fixed answer patterns, so subsequent training should place greater emphasis on exploring their reasoning capabilities. Second, our analysis points to two main factors that keep the advantage consistently zero, and thus hold *RUR* near 70% rather than at 100%: (i) nearly half of these math problems are simple enough that the model solves them reliably, and excluding such cases from

policy updates is justified because the model’s core capabilities need no further refinement there. (ii) The remaining cases are consistently answered incorrectly, which can happen for two reasons. On the one hand, some problems are too hard for the current model to solve even given a correct ground-truth label. On the other hand, the DAPO-Math-17K dataset contains some items whose labels were converted into mislabeled integers, introducing spurious ”incorrect” labels. Discarding zero-advantage responses in such cases acts as a form of implicit filtering, mitigating the risk of updating the model on noisy or mislabeled data—a robustness benefit that would carry over to any dataset with similar label noise.

Unlike GRPO and DCPO, DAPO discards responses with identical labels from the generated data, which makes a direct *RUR* comparison difficult. We therefore gauged DAPO’s training efficiency by measuring the number of generated responses and the training time needed to complete the same number of update steps. To offset the low sampling efficiency that GRPO’s low *RUR* induces, we set the generation batch size to 1–4 times the training batch size and adjusted it dynamically when training DAPO. Across all four models, DAPO required 3 to 5 times more generated responses than DCPO to reach the same number of parameter updates, at least doubling the training GPU hours without any performance gain. We further found that DAPO’s data-utilization efficiency degrades as model size grows: the fraction of responses actually used by DAPO on Qwen2.5-14B fell below 30% after about 300 steps. This scale-dependent decline indicates that DAPO’s dynamic sampling suffers from low data utilization and training efficiency, and forgoes valuable opportunities to explore model diversity—a limitation that becomes more acute precisely as models are scaled up.

In summary, DCPO marks a substantial advance over both GRPO and DAPO, markedly improving response utilization and training efficiency, making fuller use of sampled data to open up more room for exploring model diversity, and lifting performance—benefits that hold consistently across the range of models and settings examined here.

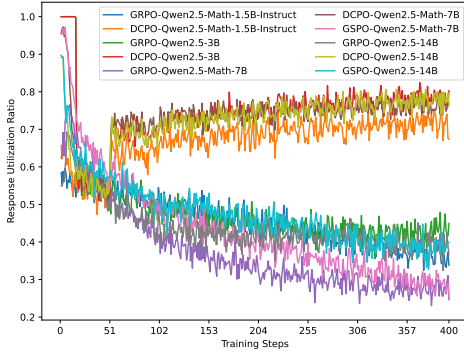


Figure 2: *RUR* progression during training, shown across models.

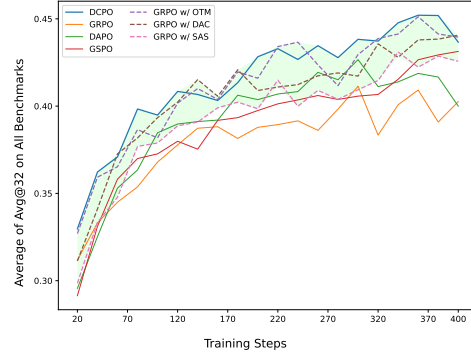


Figure 3: Ablation on the average Avg@32, using Qwen2.5-Math-7B as a representative model.

5.4 ABLATION STUDIES

We conducted an ablation study on Qwen2.5-Math-7B, taken as a representative model, to isolate the contribution of each DCPO component, using **Avg@32** as the evaluation metric because it foregrounds the robustness and stability of the learned policy distribution. For a fair comparison, each experiment alters a single component of the baseline GRPO framework while holding all other settings fixed and removing the KL-divergence term to align with DAPO, GSPO, and the full DCPO. The learning curves from 20 to 400 steps are shown in Figure 3.

The learning curves for the three individual components show that inference ability improves steadily, with the model surpassing GRPO and matching or exceeding DAPO and GSPO. Each modification thus contributes a distinct benefit, and their combination yields a substantial overall gain over the baselines—evidence that the DCPO pipeline is broadly effective for reinforcement learning tasks rather than reliant on any single ingredient. In the later stages of training the baselines plateau, whereas DCPO keeps trending upward, both at the component level and overall. The ablations confirm that every component contributes positively to training efficiency and that their synergistic integration produces cumulative gains and greater stability. Detailed results follow.

(i) **Only-Token-Mean loss (OTM)**: aggregating the loss within a single response rather than across the whole batch yields a model that outperforms the baselines, which otherwise suffer from lower data utilization and fluctuating performance. (ii) **Smoothed Advantage Standardization (SAS)**: replacing the traditional step-wise advantage with the cumulative smoothed advantage substantially improves GRPO and approaches DAPO, showing that aggregating reward statistics across training steps stabilizes optimization and lifts final performance. (iii) **Dynamic Adaptive Clipping (DAC)**: substituting a probability-adaptive clipping boundary for GRPO’s fixed one yields marked improvements; DAC alone surpasses GRPO, DAPO, and GSPO and is more stable than OTM alone. (iv) **Combination of OTM + SAS + DAC (full DCPO)**: combining all three mechanisms delivers the best and most stable results, significantly outperforming GRPO, DAPO, and GSPO. The synergy among OTM, SAS, and DAC shows that each design choice contributes uniquely to data-efficient and stable reinforcement learning, improving both the performance and the robustness of large language models generally.

Overall, the ablation confirms that each DCPO component contributes positively to performance and that their combination produces substantial cumulative gains. Taken together, the results validate the proposed mechanisms as broadly effective for improving data efficiency and stability in reinforcement learning for LLMs.

6 CONCLUSION

In this work, we set out to address limitations that run through existing policy optimization methods: restricted token-level exploration caused by fixed clipping in low-probability regions, and markedly low sample efficiency caused by standardizing rewards only within the current step. To this end we introduced **Dynamic Clipping Policy Optimization (DCPO)**, a general reinforcement learning pipeline that strengthens the reasoning capabilities of large language models through two complementary innovations: (i) a **dynamically adaptive token-level clipping** mechanism that reshapes bounds according to old-policy probabilities, enabling more efficient exploration of rare, low-probability tokens, and (ii) a **Cumulative-Smooth Advantage Standardization** that aggregates the reward distribution across optimization steps, tempering the reward randomness introduced by high-entropy sampling. Because both mechanisms depend only on quantities available to any clipping-based policy-optimization method, they are readily transferable across models and settings. Extensive experiments spanning four mathematical reasoning benchmarks and four models of differing scale and pretraining type demonstrate DCPO’s effectiveness: the DCPO-7B variant, for instance, reached an Avg@32 of 38.8 on the challenging AIME24 benchmark, outperforming GRPO (32.1) and DAPO (31.6). DCPO also raised the response utilization ratio by 28% over GRPO, doubled training efficiency relative to DAPO, and cut the token clipping ratio by an order of magnitude. Together these results indicate that DCPO substantially improves data utilization and diversity exploration in RLVR across a broad range of conditions. While our evaluation centers on mathematical reasoning, the design makes no assumptions specific to that domain, and future work will explore extending DCPO to further domains such as code generation and semantic reasoning, as well as to additional model families.

REFERENCES

- Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Naveen Sachdeva, Inderjit Dhillon, Marcel Blicstein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025.
- DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, Damai Dai, Daya Guo, Dejian Yang, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Han Bao, Hanwei Xu, Haocheng Wang, Haowei Zhang, Honghui Ding, Huajian Xin, Huazuo Gao, Hui Li, Hui Qu, J. L. Cai, Jian Liang, Jianzhong Guo, Jiaqi Ni, Jiashi Li, Jiawei Wang, Jin Chen, Jingchang Chen, Jingyang Yuan, Junjie Qiu, Junlong Li, Junxiao Song, Kai Dong, Kai Hu, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Lei Xu, Leyi Xia, Liang Zhao, Litong Wang, Liyue Zhang, Meng Li, Miaojuan Wang, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingming Li, Ning Tian, Panpan Huang, Peiyi Wang, Peng Zhang, Qiancheng Wang, Qihao Zhu, Qinyu Chen, Qiushi Du, R. J. Chen, R. L. Jin, Ruiqi

- Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, Runxin Xu, Ruoyu Zhang, Ruyi Chen, S. S. Li, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shaoqing Wu, Shengfeng Ye, Shengfeng Ye, Shirong Ma, Shiyu Wang, Shuang Zhou, Shuiping Yu, Shunfeng Zhou, Shuting Pan, T. Wang, Tao Yun, Tian Pei, Tianyu Sun, W. L. Xiao, Wangding Zeng, Wanjia Zhao, Wei An, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, X. Q. Li, Xiangyue Jin, Xianzu Wang, Xiao Bi, Xiaodong Liu, Xiaohan Wang, Xiaojin Shen, Xiaokang Chen, Xiaokang Zhang, Xiaosha Chen, Xiaotao Nie, Xiaowen Sun, Xiaoxiang Wang, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xingkai Yu, Xinnan Song, Xinxia Shan, Xinyi Zhou, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, Y. K. Li, Y. Q. Wang, Y. X. Wei, Y. X. Zhu, Yang Zhang, Yanhong Xu, Yanhong Xu, Yanping Huang, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Li, Yaohui Wang, Yi Yu, Yi Zheng, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Ying Tang, Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yu Wu, Yuan Ou, Yuchen Zhu, Yudian Wang, Yue Gong, Yuheng Zou, Yujia He, Yukun Zha, Yunfan Xiong, Yunxian Ma, Yuting Yan, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Z. F. Wu, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhen Huang, Zhen Zhang, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhibin Gou, Zhicheng Ma, Zhigang Yan, Zhihong Shao, Zhipeng Xu, Zhiyu Wu, Zhongyu Zhang, Zhuoshu Li, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Ziyi Gao, and Zizheng Pan. Deepseek-v3 technical report, 2024. URL <https://arxiv.org/abs/2412.19437>.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *arXiv preprint arXiv:2103.03874*, 2021a.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *Sort*, 2(4): 0–6, 2021b.
- Jia Li, Edward Beeching, Lewis Tunstall, Ben Lipkin, Roman Soletskyi, Shengyi Huang, Kashif Rasul, Longhui Yu, Albert Q Jiang, Ziju Shen, et al. Numinamath: The largest public dataset in ai4maths with 860k pairs of competition math problems and solutions. *Hugging Face repository*, 13:9, 2024.
- Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Understanding r1-zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*, 2025.
- MAA. American invitational mathematics examination (AIME). Mathematics Competition Series, 2024. URL <https://maa.org/math-competitions/aime>.
- OpenAI. Introducing openai o3 and o4-mini. 2025. URL <https://openai.com/index/introducing-o3-and-o4-mini/>.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. Hybridflow: A flexible and efficient rlhf framework. *arXiv preprint arXiv:2409.19256*, 2024.

- Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, et al. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for llm reasoning. *arXiv preprint arXiv:2506.01939*, 2025.
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024a.
- An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, et al. Qwen2.5-math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024b.
- Deheng Ye, Zhao Liu, Mingfei Sun, Bei Shi, Peilin Zhao, Hao Wu, Hongsheng Yu, Shaojie Yang, Xipeng Wu, Qingwei Guo, et al. Mastering complex control in moba games with deep reinforcement learning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pp. 6672–6679, 2020.
- Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Tiantian Fan, Gaohong Liu, Lingjun Liu, Xin Liu, et al. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.
- Yufeng Yuan, Qiyang Yu, Xiaochen Zuo, Ruofei Zhu, Wenyuan Xu, Jiaze Chen, Chengyi Wang, TianTian Fan, Zhengyin Du, Xiangpeng Wei, et al. Vapo: Efficient and reliable reinforcement learning for advanced reasoning tasks. *arXiv preprint arXiv:2504.05118*, 2025.
- Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in llms beyond the base model? *arXiv preprint arXiv:2504.13837*, 2025.
- Chujie Zheng, Shixuan Liu, Mingze Li, Xiong-Hui Chen, Bowen Yu, Chang Gao, Kai Dang, Yuqiong Liu, Rui Men, An Yang, et al. Group sequence policy optimization. *arXiv preprint arXiv:2507.18071*, 2025.

A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_\theta(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_\theta(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_\theta \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

Across previous methods, including GRPO and DAPO, the advantage $\hat{A}_{j,t}^i$ for token t in response j is computed by standardizing the reward R_j^i against the mean μ^i and standard deviation σ^i of the rewards of the G responses generated at step i , as shown in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

Whenever the rewards for the same prompt happen to be identical, responses with zero advantage contribute nothing to the model update, wasting generated responses—a failure mode common to methods built on this standardization scheme.

A.3.1 GRPO

Shao et al. (2024) proposed **Group Relative Policy Optimization (GRPO)**, a widely used method that first samples G responses for each query, assigns rewards R through a rule-based reward function, and estimates token-level advantages as in Equation (11). The model parameters are then updated via Equation (12), the **Sequence Level Mean loss (SLM)**:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages over all tokens in a response and then over all G responses in the batch. The $\mathbb{D}_{\text{KL}}$ penalty keeps the policy from drifting too far from a reference policy (e.g., a pre-trained model), and the fixed clipping threshold ε curbs overly large updates by limiting the change in token probability ratios. Though widely adopted, this averaging strategy inherently dilutes per-response importance, a drawback that persists wherever it is applied.

A.3.2 DAPO

Similarly, Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)**, motivated by the observation that in GRPO longer sequences exert a disproportionate influence on the gradient update. DAPO standardizes token counts across the entire batch, yielding the **Token Level Mean loss (TLM)** of Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. DAPO discards any prompt whose responses all share an identical reward and regenerates responses to maintain the batch size, a procedure that leads to severe data inefficiency and grows more costly as models scale.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping can inject high-variance noise into gradient estimates—noise that tends to accumulate over long sequences and is further amplified by the clipping operation itself. GSPO alleviates this by replacing token-wise clipping with sequence-wise clipping, which it reports to yield a more stable and effective training signal across settings.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

$$\text{where } s_i(\theta) = \exp \left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log \left(\frac{\pi_{\theta}(o_{j,t}|q)}{\pi_{\theta_{\text{old}}}(o_{j,t}|q)} \right) \right)$$

In practice, GSPO discards more than 10% of responses that carry non-zero advantages, incurring considerable data waste and reduced training efficiency. And while GRPO discards responses exhibiting sequence-level high variance, it still retains many responses with token-level high entropy, which contributes to training instability. These trade-offs, seen across methods, motivate an approach that avoids wholesale discarding.

A.4 REWARD CALCULATION

To supply a verifiable and interpretable supervision signal, we adopt a rule-based reward model that scores each generated response on both answer correctness and format compliance, as in Equation (15). A reward of +1 is assigned when both the format and the answer are correct, 0 when the format is correct but the answer is wrong, and -1 when the format is incorrect. Being rule-based, this reward scheme is simple to reproduce and readily portable to other verifiable-reward tasks.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j is the index of the response group for a given problem in the i -th training step, and the reward is calculated based on the format and correctness of the response.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Clipping the probability ratio is a crucial and broadly used technique in reinforcement learning for stabilizing policy optimization and preventing large, destabilizing parameter updates. Its importance is most acute whenever importance sampling is used to estimate expected values under a new policy from samples generated by an old one—a situation shared by a wide class of algorithms.

The expected value of a function $f(x)$ under the new probability $p(x)$ can be rewritten as an expectation under the old probability $q(x)$ through the importance-sampling weight, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x)\frac{p(x)}{q(x)}q(x) dx = \mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, its variance can be substantially inflated—a challenge common to importance sampling in general. The true variance of $f(x)$ under the new policy is given by Equation (17), whereas the variance of the importance-sampled estimator under the old policy is given by Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p}[f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x)\frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The difference between the importance-sampled variance and the true variance highlights the potential for inflation, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To mitigate this difference, a wide range of prior work—including PPO Schulman et al. (2017) and RLVR methods (DeepSeek-AI et al., 2024; Liu et al., 2025)—employs a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

A fixed clipping bound, however, inherently limits the policy’s capacity for exploration, and this is especially problematic in regions where the old policy assigns low probability mass. There the absolute bound on low probabilities is severely constrained, restricting the scope for meaningful policy updates. Although methods such as DAPO try to address this with asymmetric clipping bounds, the fundamental limitation of a non-adaptive constraint remains across such variants.

In addition, recent studies such as Yue et al. (2025) suggest that the reasoning paths exploited by RLVR models already lie largely within the sampling distribution of the base model, an observation reinforced by the minimal parameter shift seen during off-policy batch updates and reflected in the small fraction of clipped parameters. Given this stability, we propose a more broadly applicable alternative: constraining the probability ratio $r(x)$ through a dynamic-adaptive mechanism that folds the probability into the restriction, as in Equation (21), in place of the conventional fixed clipping of Equation (20).

$$|(\frac{p(x)}{q(x)} - 1)p(x)| \leq \epsilon \quad (21)$$

Substituting $p(x) = r(x)q(x)$ into the preceding inequality and expanding yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For this clipping ratio to be mathematically valid and meaningful, it must respect the fundamental constraints on probability distributions and their ratios. As shown in Equation (23), the probabilities $p(x)$ and $q(x)$ must lie between 0 and 1; consequently the probability ratio $r(x)$ must be non-negative, as must the clipping hyperparameter ϵ . These conditions hold generally, independent of the task or model.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To ensure both the validity and the practical applicability of the clipping bound derived from Equation (22), we specify the necessary conditions on its bounds as outlined in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Proceeding from Equation (22) to Equation (24), the probability ratio $r(x)$ must satisfy the following conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the clipping bound, we consider the overlapping region centered on the ratio value of 1, which minimizes the discrepancy between the two conditions and keeps the scheme consistent with the conventional fixed bound of GRPO. We then define the following conditions to guarantee the validity of the clipping bound:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, we keep the radicand non-negative by comparing it with 0 and taking the maximum, as in $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. This finally yields the following conditions, which serve as **dynamic-adaptive clipping bounds for importance sampling** and apply to any importance-sampled estimator:

$$0.5 + \frac{1}{2}\sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower clipping bounds in Inequality 27, we use distinct upper and lower clipping thresholds ϵ_{high} and ϵ_{low} so as to align the clipping bound with GRPO at specific reference points, a calibration that transfers unchanged across models.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To align our dynamic-adaptive clipping bounds with the fixed symmetric bounds at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d, we derive specific values for ϵ_{low} and ϵ_{high} by solving the equations in Equation (28).

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \\ q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases} \quad (28)$$

In common practice (e.g., GRPO) the fixed clipping threshold ϵ is set to 0.2; under this convention our clipping hyperparameters come out to approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Here the inequality $\epsilon_{low} \neq \epsilon_{high}$ serves a different purpose from the asymmetric fixed clipping used in DAPO, so the two should not be conflated.

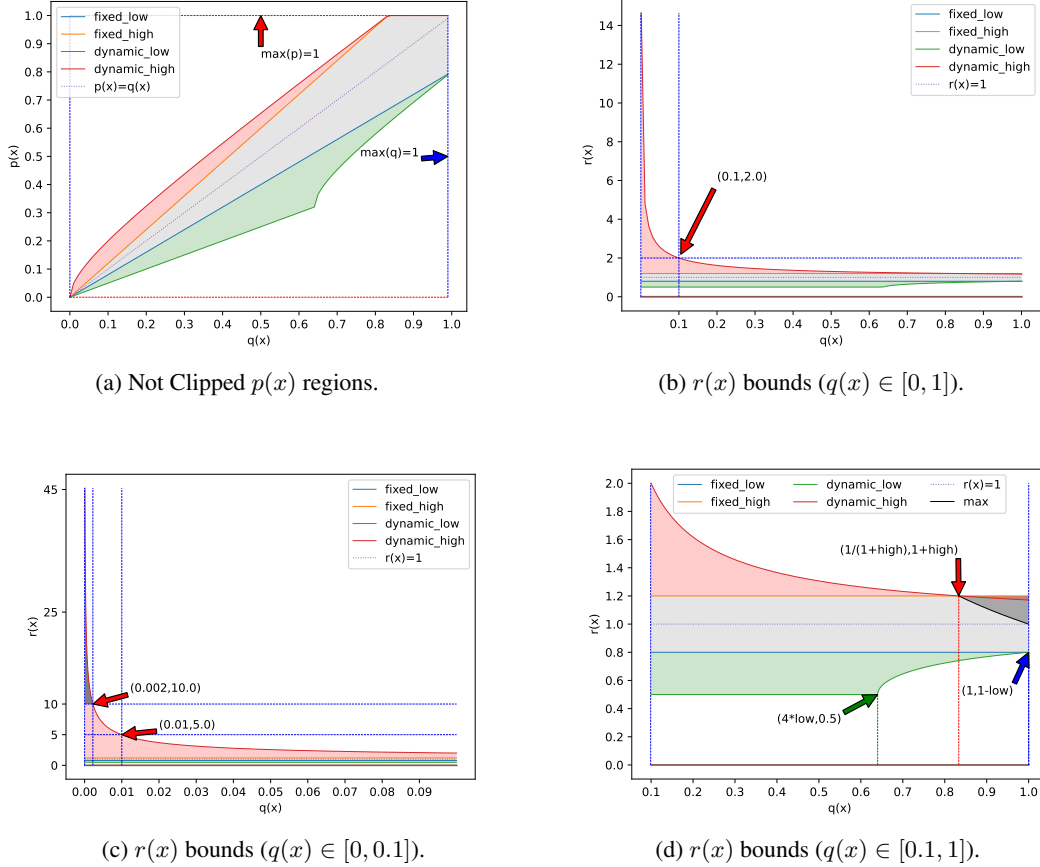


Figure 4: Comparison of clipping bounds over the full probability range. Lines show the bounds for fixed clipping ($\epsilon = 0.2$) versus dynamic-adaptive clipping ($\epsilon_{low} = 0.16, \epsilon_{high} = 0.2$).

Dynamic versus fixed clipping: Figure 4 illustrates how the bound behaves under the fixed clipping scheme used in GRPO ($[1 - \epsilon, 1 + \epsilon]$) and under our probability-aware adaptive clipping. For a token with old probability $q(x)$, the fixed bound provides a constant relative clipping width of 2ϵ , so as $q(x)$ decreases the absolute exploration space shrinks linearly (Figure 4a). The relative width under dynamic adaptation, by contrast, is inversely proportional to $q(x)$, leaving the exploration

space roughly constant across the entire probability range—behavior that holds for tokens of any prior probability.

Effect of decreasing $q(x)$: When $q(x)$ becomes small, the admissible interval $[1 - \varepsilon, 1 + \varepsilon]$ of the fixed scheme is independent of $q(x)$, so the model’s ability to explore low-probability tokens does not improve. The dynamic-adaptive bound instead expands rapidly (Figure 4b), granting a larger relative space to rare tokens—precisely the behavior required for efficient RL wherever such tokens arise.

Behavior for extremely low probabilities: Figure 4c shows the clipping interval when $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the upper bound on $r(x)$ under dynamic-adaptive clipping grows rapidly—inversely with the square root—while the fixed-clipping value stays constant. Inspired by the dual-clipping strategy (Ye et al., 2020), we enforce a hard ceiling $r_{\max} = 10$ to forestall the attendant instabilities (gradient vanishing, explosion, and the like). The gray region in Figure 4c corresponds to approximately $q(x) \leq 2 \times 10^{-3}$, where the upper limit of adaptive clipping is set to $r_{\max}$.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\varepsilon$ dynamic-adaptive clipping yields $dynamic_{low} = 0.5$, whereas fixed clipping retains the constant $fixed_{low} = 1 - \varepsilon = 0.8$. As $q(x)$ rises beyond this threshold, the dynamic-adaptive lower bound smoothly contracts toward the fixed value. Likewise, the dynamic-adaptive upper bound exceeds the fixed one whenever $q(x) < \frac{1}{1+\varepsilon_{high}}$ and merges with it for larger $q(x)$. The adaptive interval $[dynamic_{low}(q(x), \varepsilon_{low}), dynamic_{high}(q(x), \varepsilon_{high})]$ therefore widens the admissible region for rare tokens in a probability-dependent way while preserving the tightness of the fixed interval for common tokens (Figure 4d)—a single rule that adapts smoothly across the whole probability spectrum.

Positive versus negative advantage: The loss in Equation (8) contains both a positive and a negative advantage term. Under the minimum-loss operation, the ratio $r(x)$ for a positive advantage is clipped to the interval $[0, dynamic_{high}(q(x), \varepsilon_{high})]$, and for a negative advantage to $[dynamic_{low}(q(x), \varepsilon_{low}), \infty)$. Since an excessively large $r(x)$ can destabilize training, we cap the ratio for both positive and negative advantages at $r_{\max} = 10$. This precaution, inspired by Ye et al. (2020), ensures that the adaptive limits never produce excessively high importance weights in either case.

In short, for both positive and negative advantages alike, dynamic-adaptive clipping gives the model more room to explore diversity whenever rare tokens carry low probability, a benefit that generalizes across the sign of the advantage.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i denote the i -th iteration in which prompts are used to generate responses and update the model parameters. We define the following statistics for reward normalization, which apply uniformly at every step.

For the mean reward across responses, we define the group of functions given in Equation (29):

$$\left\{ \begin{aligned} \mu_{new}^i &= \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i &= \frac{1}{G * (i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total} &= \frac{1}{G * i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ &= \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{aligned} \right. \quad (29)$$

For the variance of the response rewards, we define the group of functions given in Equation (30):

$$\begin{cases}
\sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\
\sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{old}^k)^2} \\
\sigma_{total}^i = \sqrt{\frac{1}{G * i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^k)^2} \\
= \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)}
\end{cases} \quad (30)$$

To reduce resource usage when computing σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29), we rely on equivalent updates in terms of μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i rather than recomputing directly from the original rewards. This incremental formulation keeps the cost low regardless of how many steps have accumulated.

A.8 THE TEMPLATE FOR EXPERIMENTS

For all experiments in this paper we use the following template, based on the Qwen-Math template (Yang et al., 2024b), applied uniformly across every model and benchmark.

Qwen-Math Template

```

<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant

```

A.9 TRAINING SETTING

We set a symmetric clipping threshold of $\epsilon = 0.2$ for GRPO; lower/upper clipping thresholds (0.2, 0.28) together with an additional 512-token soft-punish cache and a maximum probability ratio of 10 for negative advantage for DAPO; lower/upper thresholds ($3e - 4$, $4e - 4$) for GSPO; and low/high thresholds (0.16, 0.2) for DCPO. For DCPO we preserve the same clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1+\epsilon)$ and $(q(x), r(x)) = (1, 1-\epsilon)$ as GRPO (Figure 4d) and cap the maximum probability ratio at 10 for both positive and negative advantages, in contrast to DAPO. These settings were held fixed across every model rather than tuned per model.

Training ran for 400 steps with a generation batch size of 512 and a parameter-update mini-batch size of 32. Using temperature 1 and top_p 1, we generated $G=16$ responses per problem, with a prompt maximum length of 1,024 and a response maximum length of 3,072—far shorter than the 20k maximum used in the original DAPO, which underscores that the gains do not hinge on very long generations.

We adapted our training codebase from Verl (Sheng et al., 2024) on $32 \times H20$ GPUs. Unless stated otherwise, hyperparameters matched those of DAPO. For loss computation we retained the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulation from GRPO or DAPO rather than the approximate weighting used in Verl, ensuring consistent behavior across different microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 reports performance using the average Avg@1 metric, obtained by averaging the greedy-decoding (Avg@1) results over the MATH500, AMC23, AIME24, and AIME25

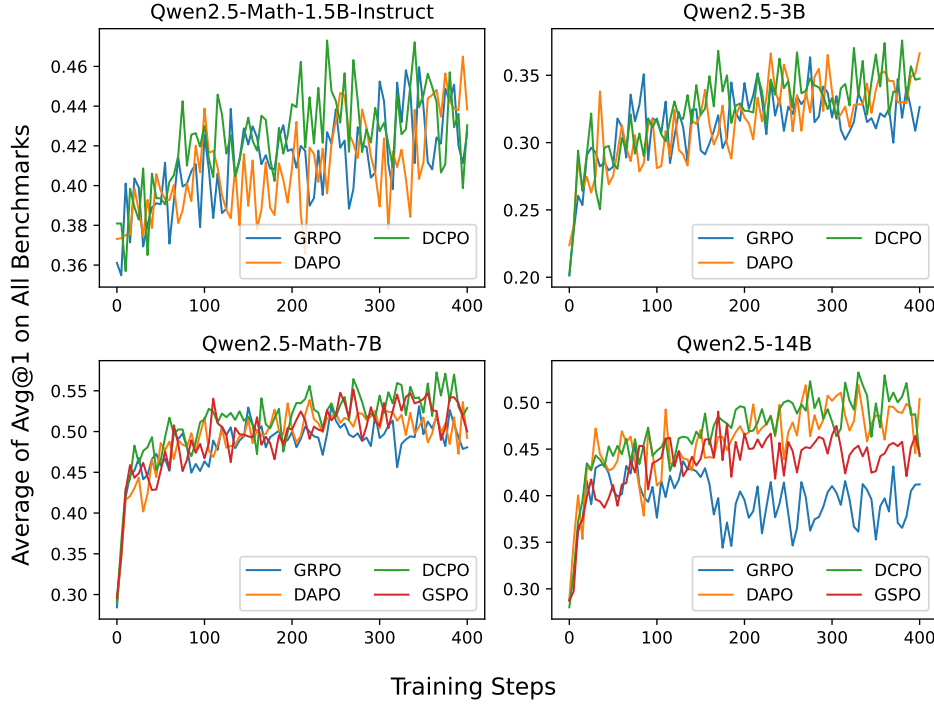


Figure 5: Avg@1 performance across benchmarks and across all four models.

benchmarks and reflecting the model’s best-case performance. The figure shows DCPO consistently outperforming GRPO and DAPO on all four models, and outperforming GSPO on the two models where GSPO was evaluated. Its overall trajectory is also steadier than the baselines’, free of the sharp drop seen with GRPO and the large fluctuations seen with DAPO. Observed uniformly across the suite, this behavior suggests that DCPO elicits stronger reasoning capabilities during training and leaves more headroom for the model to reach still better performance.

Avg@32 Trend Figure 6 plots the Avg@32 performance of DCPO alongside GRPO, DAPO, and, where available, GSPO across model sizes on the AMC23, AIME24, and AIME25 benchmarks, clearly bringing out DCPO’s exceptional robustness and stability. GRPO, by contrast, fluctuates heavily and occasionally collapses; under GRPO the Qwen2.5-14B model even suffers a pronounced drop after roughly 170 steps. Across every evaluated model, from 1.5B to 14B, DCPO’s trajectory trends consistently upward with minimal fluctuation, sustaining the fastest early growth and holding its upward course throughout. Compared with DAPO, moreover, DCPO reaches superior performance while halving training time and resources. Taken together, these results demonstrate that dynamic-adaptive clipping and smooth advantage standardization improve response-level and label-level exploration of diversity effectively and robustly across the full range of model scales tested.

Entropy Trend Figure 7 shows GRPO undergoing a sharp entropy decay after an initial warm-up phase and quickly settling into a flat, low-entropy range. This entropy collapse reflects a rapid loss of policy randomness that shrinks the diversity of sampled responses and thereby limits subsequent policy optimization.

The behavior of DAPO and DCPO, in turn, depends on the model. On the instruction-tuned model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO all follow similar declining trajectories. On the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), the trends diverge markedly: DAPO holds the highest entropy, rising clearly but with large fluctuations. On Qwen2.5-Math-7B, GSPO’s entropy is close to DCPO’s and their average performance is comparable, whereas on Qwen2.5-14B GSPO drifts downward with fluctuations and performs poorly. DCPO, by contrast, reaches a moderate entropy level, with a markedly smoother curve confined to a bounded band

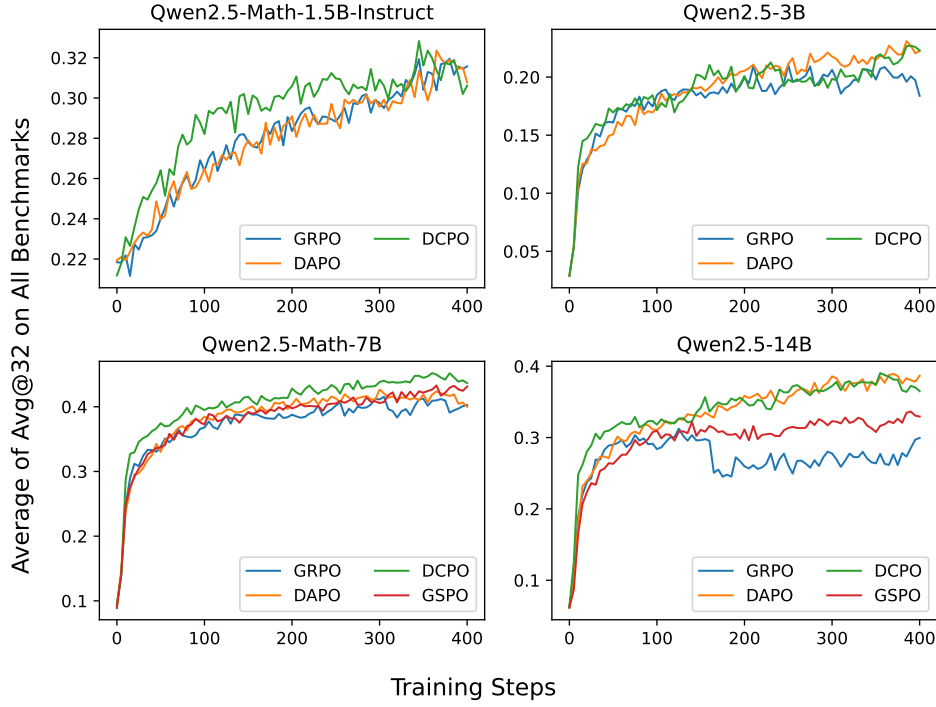


Figure 6: Avg@32 performance across benchmarks for all model scales.

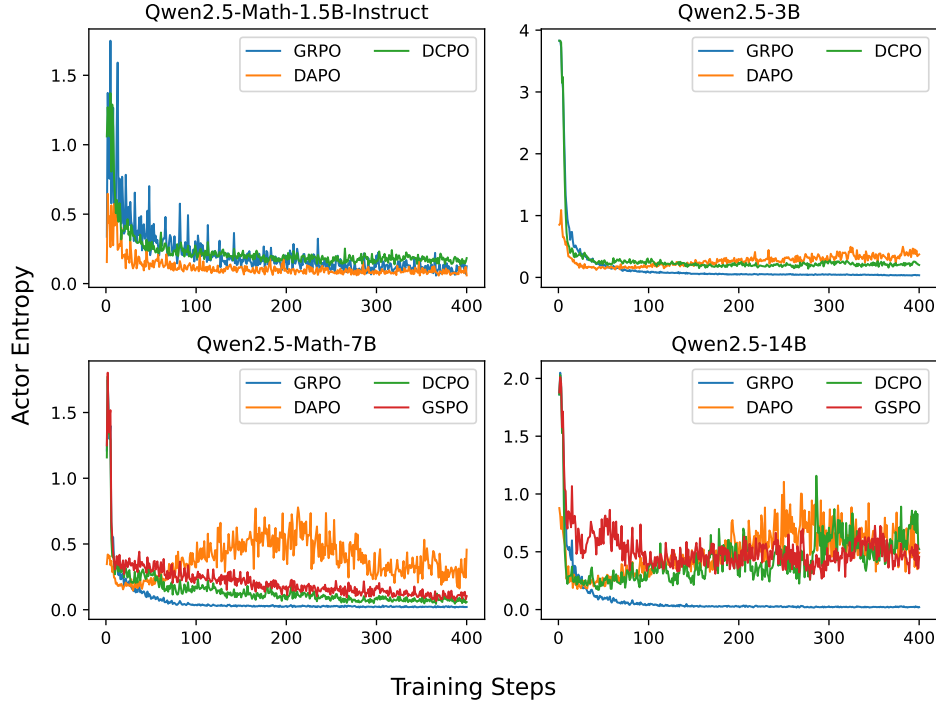


Figure 7: Policy entropy of the models during training, across all four models.

between GRPO’s low-entropy region and DAPO’s high-variance region—a pattern that recurs across the base models.

These observations point to a U-shaped relationship between entropy and training performance that appears to hold generally: too little entropy starves the learner of exploratory signal and induces premature convergence, while excessively high or unstable entropy undermines training stability. By keeping entropy within a moderate, well-balanced band, DCPO fosters both convergence and sufficient exploration, and this benefit becomes more pronounced as model capacity grows. Its entropy-regulation mechanism thus offers a plausible explanation for DCPO’s superior inference performance and for its more robust policy learning across models of widely varying size.